Unit IV

Data Analysis

Descriptive Statistics

It is a summary statistics that quantitatively describes or summarizes features from a collection of information.

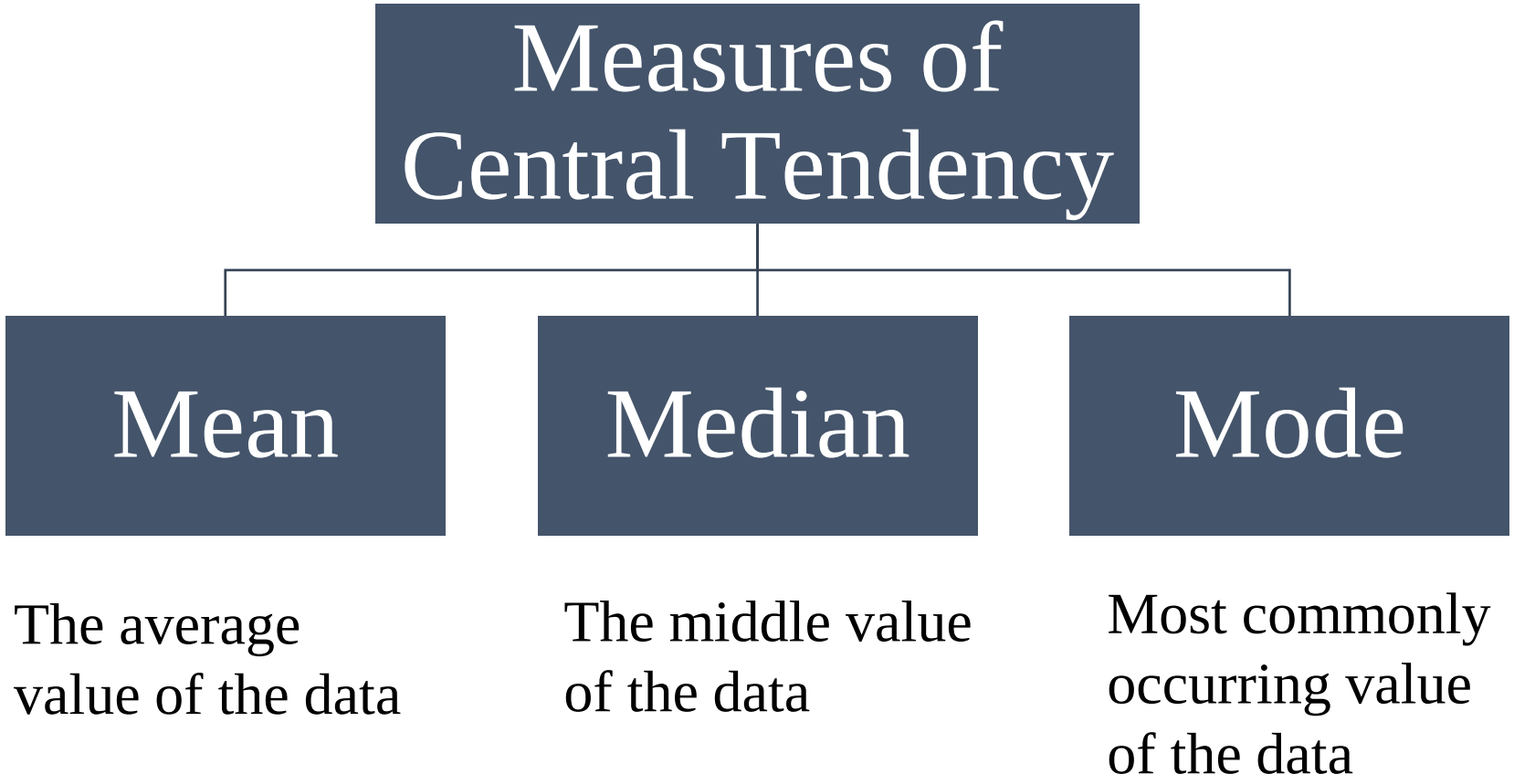
In quantitative research, after collecting data, the first step of statistical analysis is to describe characteristics of the responses, such as the average of one variable (e.g., age), or the relation between two variables (e.g., age and creativity).

Central Tendencies
Measures of Variation or Dispersion

Measures of Central Tendency

Introduction

A measures of central tendency is a statistical measure that determines a single value that accurately describes the center of the distribution and represents the entire distribution of data.



Classification of Data

Example : Consider marks of 10 students (out of 25)

23,10,12,23,20,12,20,15,15,25

(Ungrouped Data)

Ungrouped Frequency Distribution	
23	2
10	1
12	2
20	2
15	2
25	1

Grouped Frequency Distribution	
Class	
0-5	0
5-10	0
10-15	3
15-20	2
20-25	5

Mean

- The mean is the arithmetic average
- Most common measure
- It is simply the sum of the values divided by the total number of items in the data set.

$$\bar{x} = \frac{\sum x}{n} = \frac{\text{the sum of the data pts}}{\text{the number of data pts}}$$

WITHOUT FREQUENCY DATA

$$\bar{x} = \frac{\sum x}{n}$$

FREQUENCY DATA

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i}$$

Median

- The median is the middle value of the data set
- It is the value that splits the dataset in half.
- It does not affected by extreme values.

For Ungrouped data:-

If X_1, X_2, X_3, ..., X_n are n observations arranged in ascending order of magnitude.

$$Median = \begin{cases} X_{(n+1)/2} & if\ n\ is\ odd \\ \dfrac{X_{n/2} + X_{(n/2)+1}}{2} & if\ n\ is\ even \end{cases}$$

For Grouped data:-

$$Median = l + \frac{\left(\frac{n}{2} - c\right)h}{f}$$

l = lower limit of class interval where the median occurs
f = Frequency of the class where median occurs
h = Width of the median class
C= Cumulative frequency of the class preceding the median class

Mode

The **mode** is the most frequent number in a collection of data.

Ex. A: 3, 10, 8, 8, 7, 8, 10, 3, 3, 3

The mode of the above example is 3, because 3 has a frequency of 4.

Ex. B: 2, 5, 1, 5, 1, 2

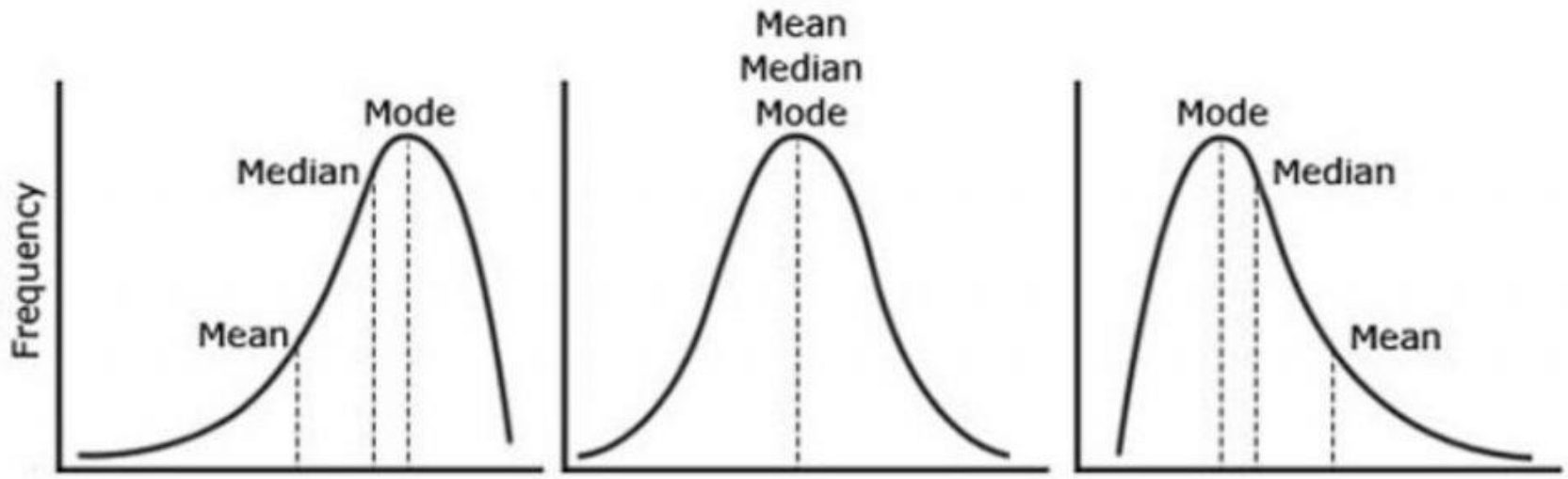
This example has no mode because 1, 2, and 5 have a frequency of 2.

Ex. C: 5, 7, 9, 1, 7, 5, 0, 4

This example has two modes 5 and 7. This is said to be bimodal.

For Grouped data:-

$$Mode = l + \frac{f_m - f_1}{2f_m - f_1 - f_2} \cdot h$$



Negatively-Skewed Symmetrical Positively-Skewed

- In symmetrical distributions, the median and mean are equal
  - For normal distributions, mean = median = mode
- In positively skewed distributions, the mean is greater than the median
- In negatively skewed distributions, the mean is smaller than the median

Measures of Dispersion

Why Measure of Dispersion?

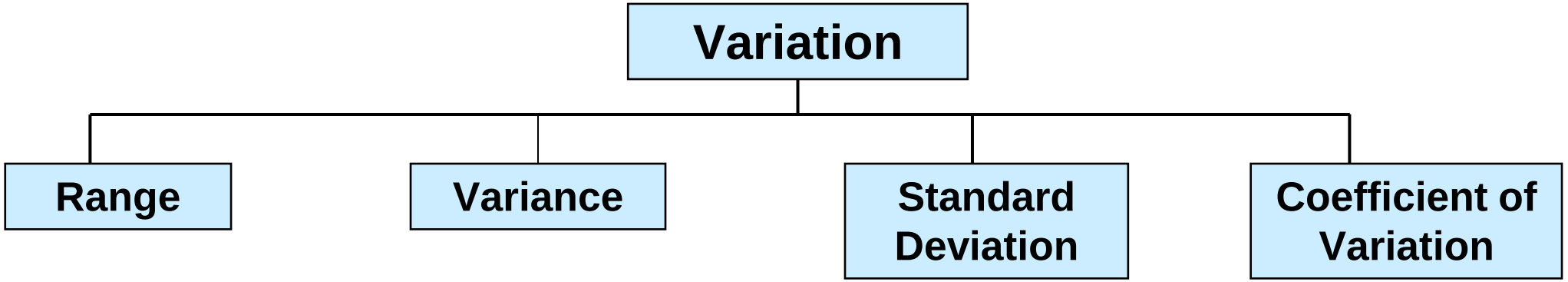
What is Measure of Dispersion?

Dispersion refers to the extent to which a set of data is spread out, or dispersed from the 'average'.

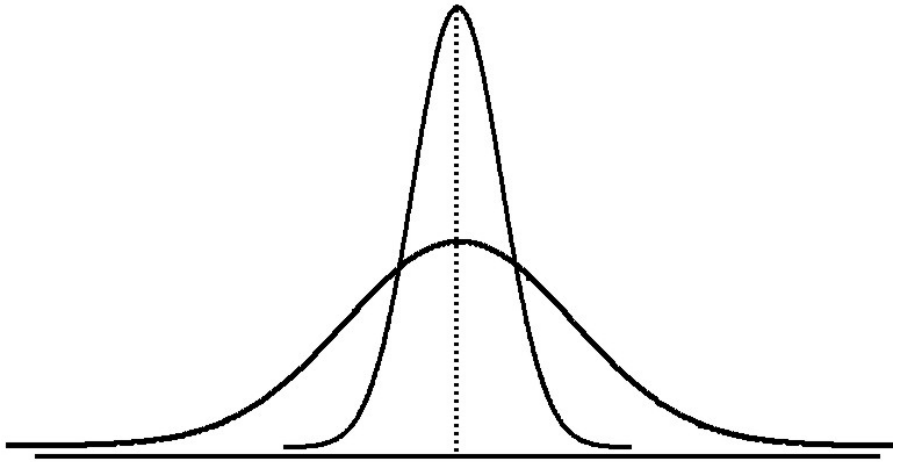
Why Measure of Dispersion?

- Serve as a basis for the control of the variability.
- To compare the variability of two or more series.
- Facilitate the use of other statistical measures.
- Determine the reliability of an average.

Measures of Dispersion



Measures of variation give information on the **spread** or **variability** or **dispersion** of the data values.



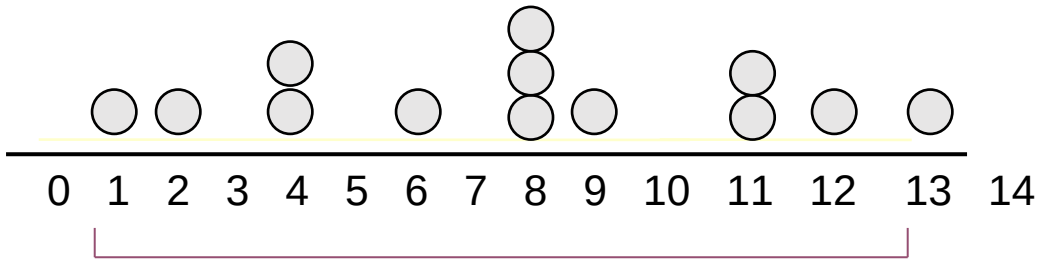
Same centre, different variation

The Range

- Simplest measure of dispersion
- Difference between the largest and the smallest values:

$$\text{Range} = X_{\text{largest}} - X_{\text{smallest}}$$

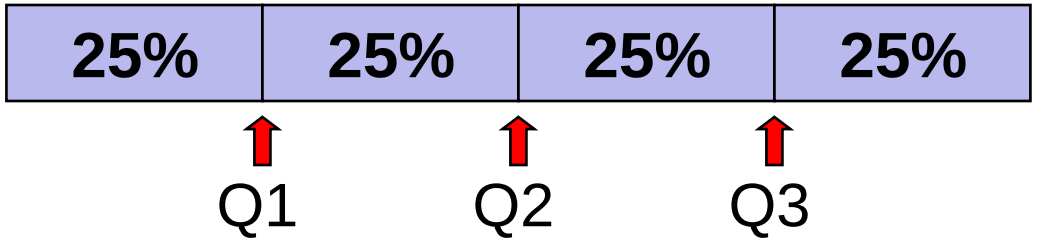
Example:



Range = 13 - 1 = 12

Quartile Measures

Quartiles split the ranked data into 4 segments with an equal number of values per segment



- The first quartile, Q_1, is the value for which 25% of the observations are smaller and 75% are larger
- Q_2 is the same as the median (50% of the observations are smaller and 50% are larger)
- For Q_3 , Only 25% of the observations are greater than the third quartile

Variance

Variance is the average (approximately) of squared deviations of values from the mean

For Ungrouped Data:

$$\sigma^2 = \frac{\sum_{i=1}^{n} \left(x_i - \bar{x}\right)^2}{n}$$

Where $\bar{x}$ = arithmetic mean

n = sample size

x_i = ith value of the variable X

For Grouped Data:

$$\sigma^2 = \frac{\sum_{i=1}^{n} x^2 f}{n} - \bar{x}^2$$

Where $\bar{x}$ = arithmetic mean

n = sample size

x_i = ith value of the variable X

f = frequency

Standard Deviation

- Most commonly used measure of variation
- Shows variation about the mean
- Is the **square root of the variance**
- Has the same units as the original data

For ungrouped data:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2}{n}}$$

For Grouped Data:

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} x^2 f}{n} - \bar{x}^2}$$

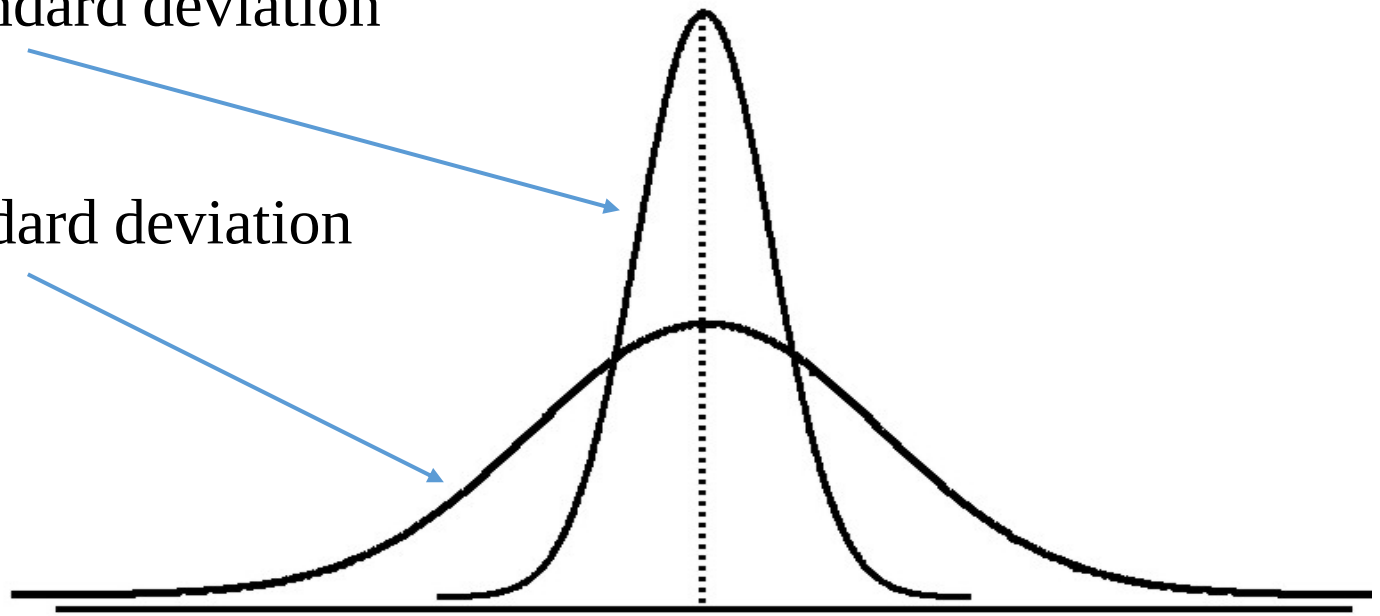
Summary Characteristics

- The **more** the data are spread out, the **greater** the range, variance, and standard deviation.
- The **less** the data are spread out, the **smaller** the range, variance, and standard deviation.
- If the values are all the same (no variation), all these measures will be zero.
- **None of these measures are ever negative.**

Summary Characteristics

Smaller standard deviation

Larger standard deviation



Coefficient of Variation

- Measures **relative variation**
- Always in percentage (%)
- Shows **variation relative to mean**
- Can be used to compare the variability of two or more sets of data measured in different units.
- Can be used to compare two distributions directly to see which has more dispersion because it does not depend on units of the distribution.

Here S is Standard Deviation, $\bar{X}$ is Mean

$$CV = \left(\frac{S}{\bar{X}} \right) \times 100$$

Coefficient of Skewness

Coefficient of skewness(β_1)

$$\beta_1 = \frac{\mu_3^{\,2}}{\mu_2^{\,3}}$$

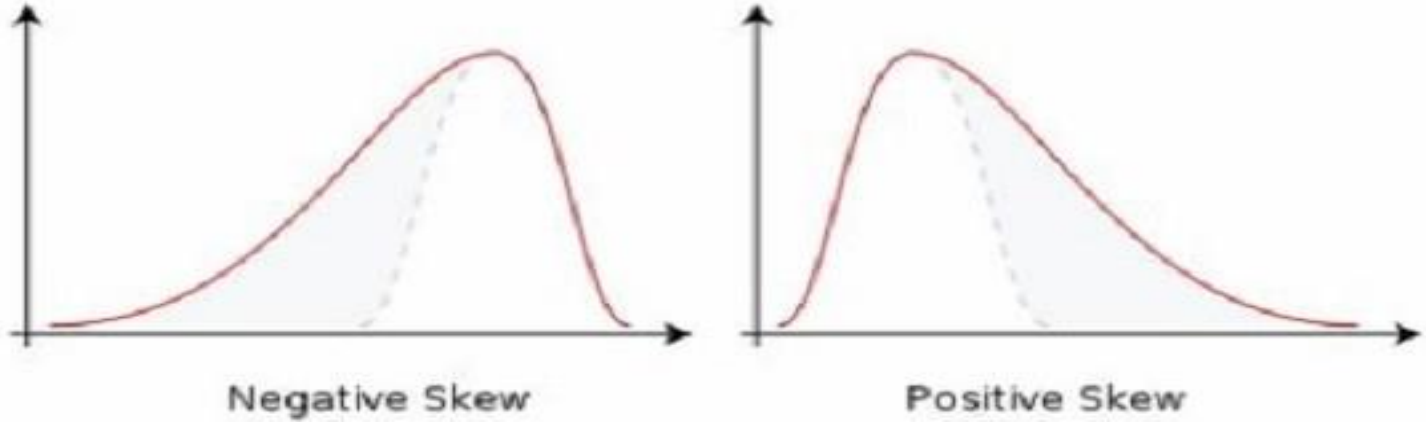
1. If $\beta_1 > 0$ then distribution is called Positive Skew Distribution
2. If $\beta_1 < 0$ then distribution is called Negative Skew Distribution
3. If $\beta_1 = 0$ then distribution is called Normal Distribution

Moments, Skewness and Kurtosis

Skewness : It gives imbalance and asymmetry from mean of a data distribution

There are two types of skewness

1. **Positive Skewness** : If the frequency curve of a distribution has a longer tail to the right of the central maximum than to the left, the distribution is said to be skewed to the right or to have positive skewed. In a positive skewed distribution, Mean > Median > Mode
2. **Negative Skewness** : If the frequency curve has a longer tail to the left of the central maximum than to the right, the distribution is said to be skewed to the left or to have negative skewed. In a negatively skewed distribution, Mode > Median > Mean.



Coefficient of Skewness

Positive Skewness in R

The asymmetry of data distributions where the tail extends towards higher values is known statistically as positive skewness. It is a crucial metric in a number of disciplines, including data analysis, social sciences, finance, and economics. The definition, computation techniques, interpretation, and applications of positive skewness.

Negatively Skewness in R

Left-skewed distributions, commonly referred to as negatively skewed distributions, are statistical notions that describe asymmetrical data distributions with a tail that slopes downward. In a number of disciplines, including finance, economics, social sciences, and data analysis, it is crucial to comprehend negatively skewed data.

Coefficient of Skewness

Zero Skewness in R

A statistical notion called zero skewness, commonly referred to as symmetry, defines data distributions that are balanced and have equal probability on both sides of the mean. It is a fundamental metric used in many disciplines, such as data analysis, economics, social sciences, and finance.

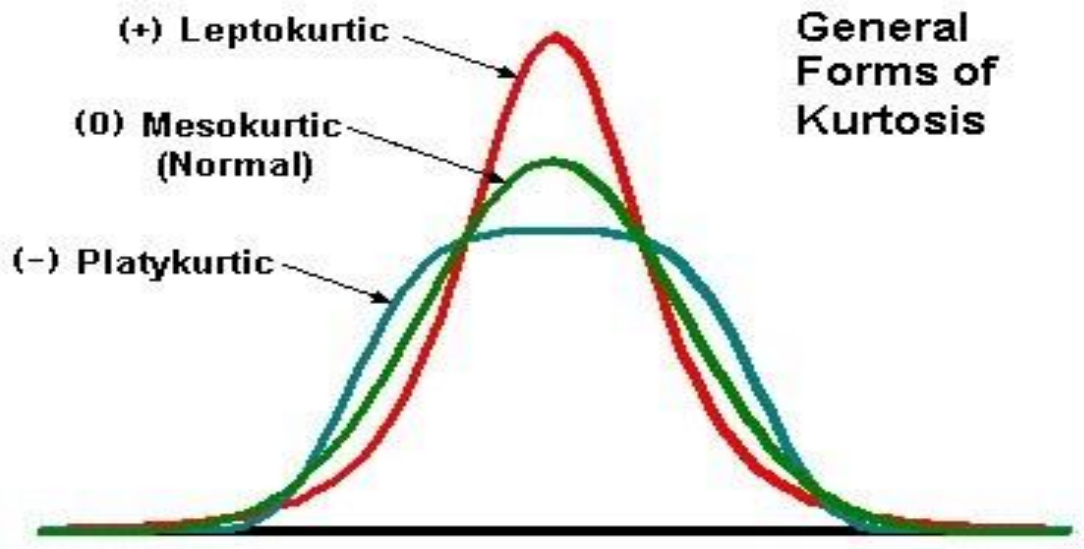
- Variance tells us about the amount of variability while skewness gives the direction of variability.

Kurtosis

Kurtosis : Kurtosis is a measure of peakness of a distribution relative to the normal distribution(or symmetrical distribution)

- A distribution having a relatively high peak is called leptokurtic $(\beta_2 > 3)$.
- A distribution which is flat topped is called platykurtic $(\beta_2 < 3)$.
- The normal distribution which is neither very peaked nor very flat-topped is also called mesokurtic $(\beta_2 = 3)$.

Coeff. of Kurtosis $(\beta_2) = \frac{\mu_4}{{\mu_2}^2}$



Kurtosis

Kurtosis expresses the vertical upward or downward pull of data-based data distribution near the mean and its relative standard deviation. This also indicates the presence of outliers based on the type of either-sided tails (flatter, heavier).

The presence of extreme lower or higher values (outliers) in datasets plays a vital role in pulling the distributions on either side.

Kurtosis is used to understand the shape of a distribution and identify whether it deviates from a normal distribution. It helps to detect outliers, assess the risk of extreme events, and evaluate the potential for high or low returns in investments or statistical analyses.

Coefficient of Correlation

Correlation:

To measure linear relationship between two variables, Karl Pearson developed a formula called correlation coefficient.

Coefficient of Correlation between two variables x and y denoted by $r(x,y)$ and defined as

$$r(x,y) = \frac{cov(x,y)}{\sigma_x \sigma_y}$$

Where

1. Ungrouped Data

$$cov(x,y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

$$cov(x,y) = \frac{1}{n}\sum_{i=1}^{n} x_i y_i - \bar{x}.\bar{y}$$

$$\sigma_x^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - (\bar{x})^2 \qquad \sigma_y^2 = \frac{1}{n}\sum_{i=1}^{n} y_i^2 - (\bar{y})^2$$

2. Ungrouped Frequency Distribution

$$cov(x,y) = \frac{1}{N}\sum_{i=1}^{n} f_i(x_i - \bar{x})(y_i - \bar{y})$$

$$cov(x,y) = \frac{1}{N}\sum_{i=1}^{n} f_i x_i y_i - \bar{x}.\bar{y}$$

$$\sigma_x^2 = \frac{1}{N}\sum_{i=1}^{n} f_i x_i^2 - (\bar{x})^2 \qquad \sigma_y^2 = \frac{1}{N}\sum_{i=1}^{n} f_i y_i^2 - (\bar{y})^2$$

Analytical tools

Structure of data

- Symmetricity
- Skewness
- Kurtosis, etc.

Relationship in data

- Correlation coefficient
- Correlation ratio
- Linear regression
- Non linear regression, etc.

Graphics of data

- Boxplot
- Frequency curve
- Histogram
- Line graph
- Barplot, etc

Flexibility of R

R is very versatile and provides us variety of tools, packages and commands for a single term of descriptive statistics.

Here, we explained individual command for each concept of statistics and finally gave a consolidated command to carry all operation with single command

Let us drill the command in R-studio